

Supplementary Material: Bundle Neural Networks for Functional Connectomes

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I. ANALYSIS CONTRACT

The manuscript is built from the locked Step 13 evidence package. SHA-256 digests bind the claim ledger, result snapshot, artifact manifest, and paper asset manifest to the build. If any bound input changes, the builder stops before generating tables or numeric macros. This supplement contains the pre-specified secondary summaries. It adds no model fitting, participant exclusions, hyperparameter selection, or confirmatory inference.

The results apply to the analyzed ABIDE-I C-PAC filtered AAL-116 pipeline, which omits global-signal regression and uses positive-edge graphs with connectivity-profile node features. Clinical utility, anatomical or causal connectivity, biological bundle geometry, and BuNN performance outside this setting were not tested.

II. REPRESENTATION ENDPOINTS

Learned-BuNN embeddings may occupy different node-specific frames. Before any cross-node diagnostic, we mapped them into a common learned frame. The primary endpoint was normalized effective rank at layer 2, measured against each operator's matched density-zero anchor. Secondary diagnostics included dispersion, cosine similarity, invariant edge-transport distance, and the rank change from encoder to layer 2.

TABLE S1
BUNN-MINUS-GCN MATCHED-ANCHOR REPRESENTATION CONTRASTS AT LAYER 2.

Endpoint	Estimate	95% site-bootstrap CI
Normalized effective rank	-0.0072	[-0.0131, -0.0010]
Normalized dispersion	-0.0417	[-0.0523, -0.0306]
Mean pairwise cosine similarity	0.0415	[0.0319, 0.0507]
Invariant edge-transport distance	-0.6436	[-0.8205, -0.4720]
Encoder-to-layer-2 effective-rank change	-0.0056	[-0.0144, 0.0048]

III. ROBUSTNESS SUMMARIES

Each held-out site formed one generalization unit, so the confirmatory summary gave every site equal weight. Participant-weighted and site-median summaries were pre-listed sensitivity analyses and did not replace the primary aggregation.

TABLE S2
PRE-LISTED ALTERNATIVE SUMMARIES. EQUAL-SITE MEANS REMAIN CONFIRMATORY.

Contrast	Summary	Estimate
BuNN - GCN curve	equal site mean	-0.00958
BuNN - GCN curve	participant weighted mean	0.00297
BuNN - GCN curve	median site difference	-0.00439
BuNN - elastic net	equal site mean	-0.05516
BuNN - elastic net	participant weighted mean	-0.04446
BuNN - elastic net	median site difference	-0.04644
BuNN - GCN effective-rank change	equal site mean	-0.00719
BuNN - GCN effective-rank change	participant weighted mean	-0.01094
BuNN - GCN effective-rank change	median site difference	-0.00917

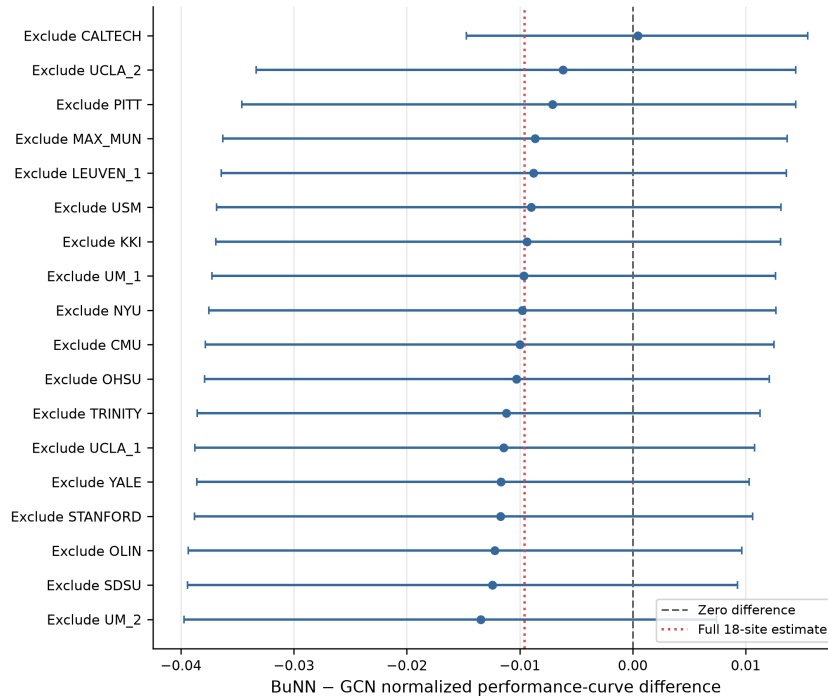


Fig. S1. Leave-one-site-out influence on the primary BuNN-minus-GCN curve contrast. The figure includes every pre-listed site exclusion.

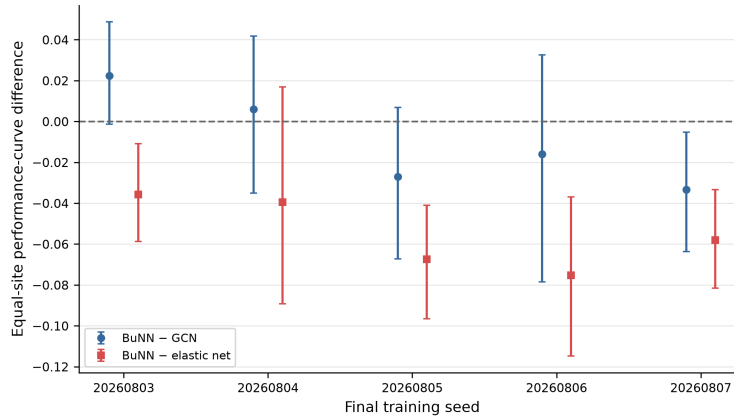


Fig. S2. Curve contrasts for each final seed. Intervals resample held-out sites within a fixed seed; these results assess robustness and do not replace the primary analysis.

For the influence analysis, we recomputed each named contrast after removing one outer site at a time. We also recomputed the primary curve contrast separately for every fixed final seed. These analyses measure sensitivity to sites and optimization while leaving the primary estimand unchanged.

IV. REPRODUCIBILITY AND INTEGRITY

The repository records the participant manifest, technical exclusion rules, connectome construction, held-out-site splits, operator contracts, tuning budgets, final seeds, source hashes, run audits, and aggregate result artifacts. The build reads the locked result snapshot and writes the LaTeX macros and tables, so manuscript values are not copied by hand. Tests check the input hashes, deterministic generation, claim coverage, bibliography resolution, and the absence of manually duplicated primary results in the narrative source.